TINA YUTING HUANG, PH.D.

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RESEARCH PROFILE

Interdisciplinary scientist developing programmable drug-delivery systems at the interface of **microfluidics, biomaterials, mRNA therapeutics, and machine learning**. Research integrates automated microfluidic manufacturing, quantitative materials characterization, simulation, and data-guided design to engineer thermally stable and sustained-release therapeutics.

ACADEMIC & RESEARCH APPOINTMENTS

Postdoctoral Associate, *Koch Institute for Integrative Cancer Research, Massachusetts Institute of Technology* 06.2023–Present
Mentors: Robert S. Langer and Ana Jaklenec

Researcher, *Harvard University* 06.2022–06.2023
Advisor: David A. Weitz

EDUCATION

Ph.D., Applied Physics, *Harvard University, School of Engineering and Applied Sciences* COMPLETED IN 2022
Thesis advisor: David A. Weitz | Thesis: "Microfluidic Production and Characterization of Lipid Vesicles, Polymer Vesicles, and Hybrid Polymer/Lipid Vesicles"

A.B., Physics; Certificate in Engineering Physics,
Princeton University

RESEARCH EXPERIENCE

MIT – Lead Research Programs

- **Machine learning–optimized core–shell microcapsules for sustained IL-10 mRNA immunotherapy in melanoma.** Developed programmable PLGA microcapsules for sustained mRNA–LNP delivery, achieving improved tumor suppression with reduced dose and administration frequency while enhancing systemic immune responses.
- **Multimodal machine learning for thermostable, single-dose COVID-19 vaccination.** Developing a framework integrating confocal imaging, COMSOL simulations, and in vitro transfection data to optimize hydrogel-core/PLGA-shell microparticles that reproduce multidose vaccine exposure through a single administration.
- **Machine-learning-enabled mRNA sequence design.** Developing a high-throughput IVT/IVTT and machine-learning platform to design and screen mRNA sequence libraries for improved translation efficiency, expression magnitude and duration, and thermal stability.
- **Microparticle-integrated microneedles for programmable drug delivery.** Integrating controlled-release microparticles into microneedles for room-temperature-stored vaccines with minimally invasive, programmable delivery.

Mentored Undergraduate Research

- **Hydrogel-based thermal stabilization of mRNA–LNP via microfluidic encapsulation (Lauren Levy, senior thesis).** Developed gel-phase encapsulation strategies that enhance mRNA–LNP thermal stability. Mentee received the MIT Bioengineering Merck Prize and Koch Institute Marcella O'Grady Boveri Medal.
- **Machine learning-guided design of acid-stable lipid nanoparticles (Chloe Tan, Emma Wang).** Integrated machine learning with high-throughput formulation screening to optimize LNP stability under acidic conditions.
- **Additional mentored projects:** fluorescence-based high-throughput screening of LNP stability (Kyle Taylor, Chloe Tan); light-triggered, magnetically guided shape-changing polymer–LNP systems for active mRNA delivery (Aiden Sanxhaku).

RESEARCH EXPERIENCE, CONTINUED

Selected Collaborative Research

- SMART: Smart Microneedle Array for Responsive Therapy – AI-assisted microneedle patch for programmable drug release (submitted); contributed simulation.
- TIMED: Temporal intervention with microparticle encapsulation and delivery – programmed release for post-myocardial infarction therapy. Cell Biomaterials. 2026;2(1); contributed simulation.
- Programmable drug-eluting coronary stent; contributed simulation and PLGA SEAL particles for manuscript in preparation.
- Engineered ginger exosomes for nose-to-brain delivery in glioblastoma therapy; contributed simulation for manuscript in preparation.

Harvard – Microfluidics, Biomaterials, and Vesicle Mechanics

- Developed microfluidic methods to produce lipid, polymer, and hybrid polymer–lipid vesicles with controlled membrane composition and asymmetry; work led to multiple peer-reviewed publications.
- Designed and built quantitative biomechanical instrumentation, including micropipette aspiration and image-analysis tools, for real-time characterization of cell and vesicle deformation and membrane mechanics.
- Collaborated across departments, co-authored an NSF-funded grant proposal, and mentored undergraduate, visiting master's, and junior graduate researchers.

SELECTED PEER-REVIEWED PUBLICATIONS

- **Huang Y**, Manafirad A, Matoori S, Arriaga LR, Sun S, Chen A, Yang X, Dinsmore AD, Mooney DJ, Weitz DA. Mechanical performance of hybrid polymer–lipid vesicles with leaflet asymmetry engineered using microfluidics. *Proceedings of the National Academy of Sciences*. 2026;123(10):e2516407123.
- Arriaga LR+, **Huang Y+,** Kim SH, Aragones JL, Ziblat R, Koehler SA, Weitz DA. Single-step assembly of asymmetric vesicles. *Lab on a Chip*. 2019;19(5):749–756.
- **Huang Y**, Kim SH, Arriaga LR. Emulsion template vesicles with symmetric or asymmetric membranes. *Advances in Colloid and Interface Science*. 2017;247:413–425.
- Michelon M+, **Huang Y+,** de la Torre LG, Weitz DA, Cunha RL. Single-step microfluidic production of W/O/W double emulsions as templates for β-carotene-loaded giant liposomes formation. *Chemical Engineering Journal*. 2019;366:27–32.
- Chen X, Chen Y, **Huang Y,** Zou L, Liu C, McClements DJ, Liu W. Hybrid bionanoparticle-stabilized Pickering emulsions for quercetin delivery: effect of interfacial composition on release, lipolysis, and bioaccessibility. *ACS Applied Nano Materials*. 2019;2(10):6462–6472.
- Levy L†, Tan C†, **Huang Y,** Yang X, Langer R, Jaklenec A. Microneedle patch-based delivery of mRNA vaccines: opportunities for next-generation immunization. *AAPS Open*. 2026;12(1):49.
- Tan C†, Levy L†, **Huang Y,** Jaklenec A. RNA Delivery Technologies: Industrial Applications and Emerging Innovations.
- Xia J, Yuan Y, Wu H, **Huang Y,** Weitz DA. Decoupling the effects of nanopore size and surface roughness on the attachment, spreading and differentiation of bone marrow-derived stem cells. *Biomaterials*. 2020;248:120014.

+ Equal contribution. † Undergraduate mentee.

SELECTED SUBMITTED / COLLABORATIVE MANUSCRIPTS

- Yang X, Zhang L, Song H, Gao W, **Huang Y,** et al. SMART: Smart Microneedle Array for Responsive Therapy – An AI-assisted microneedle patch-based programmable drug release platform. Submitted.
- Wang EY, Calle EA, Ying B, Eshaghi B, Zhang L, Yang X, Lin SQ, Han J, Backx AG, **Huang Y,** Mursalova S. TIMED: Temporal intervention with microparticle encapsulation and delivery—A programmed release system for post-myocardial infarction therapy. *Cell Biomaterials*. 2026;2(1).

MENTORING & TEACHING

Mentored 15+ undergraduate and junior researchers at Harvard and MIT. Mentees have produced undergraduate first-authored papers and earned the MIT Undergraduate Research Symposium in **Bioengineering Best Poster Award**, the MIT Bioengineering **Merck Prize** for best senior thesis, and the Koch Institute **Marcella O'Grady Boveri Medal** for best senior thesis.

- Mentoring approach emphasizes quantitative experimental design, independent troubleshooting, scientific writing, presentation, and progressive research ownership.

TECHNICAL EXPERTISE

Microfluidics & fabrication: soft lithography, emulsion microfluidics, core–shell microparticles, microneedles | Biomaterials & drug delivery: mRNA–LNPs, hydrogels, polymers, vesicles, controlled release | Characterization: SEM, TEM, fluorescence imaging, flow cytometry, AFM, micropipette aspiration | Computation: physics-informed machine learning, morphology-based AI, COMSOL/multiphysics, MD and CFD simulation | Biological methods: in vivo animal models, single-cell sequencing

PATENTS

- Lipid-polymer microparticles for thermostable, pulsatile mRNA-LNP vaccines. U.S. Patent Application 63/733,652, filed Dec. 13, 2024.
- Stabilization of mRNA-LNP using antioxidants. U.S. Patent Application 63/733,778, filed Dec. 13, 2024.

SELECTED CONFERENCE PRESENTATIONS

- Huang Y, et al. "Lipid-Polymer Microparticles (mLPM) for Thermostable, Pulsatile mRNA-LNP Vaccines." ACS Fall 2025 Meeting, Aug. 19, 2025; Abstract ENFL 249.
- Huang Y, Zhang R, Lin S, Langer RS, Jaklenec A. "Thermostable pulsatile mRNA vaccination using mRNA-LNPs inside lipid-polymer microparticles (mLPM)." 16th US-Japan Symposium on Drug Delivery Systems, Lahaina, HI, Dec. 2024.

SELECTED AWARDS & HONORS

- Koch Institute Convergence Scholar Award; Koch Institute Marlena Felter Bradford Research Travel Award.

RESEARCH LEADERSHIP & SERVICE

- Mentored undergraduate and junior researchers across independent and collaborative projects in microfluidics, mRNA delivery, biomaterials, and machine-learning-guided formulation design.
- Contributed quantitative simulation, materials design, and microfluidic expertise to cross-disciplinary collaborations spanning cardiovascular drug delivery, microneedles, and brain delivery.